

DONGSHENG, China

WMO: 535430

Lat: 39.8155N

Lon: 110.0135E

Elev: 4787

StdP: 12.33

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.7	1.0	-20.7	2.1	7.5	-17.1	2.6	10.9	18.2	18.5	15.7	17.1	5.4	290	0.503

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.9	85.8	60.0	83.5	59.6	81.1	59.1	67.5	76.4	65.8	74.6	64.4	73.5	7.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.8	110.1	71.0	63.1	103.3	70.8	61.1	96.1	69.4	35.0	76.4	33.5	74.8	32.3	73.7	72.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	16.7	14.6	12.8	DB	-7.7	90.0	4.5	3.1	-10.9	92.2	-13.5	94.1	-16.0	95.8	-19.3	98.1
(5)				WB	-8.7	69.6	4.3	1.8	-11.8	70.9	-14.3	71.9	-16.7	73.0	-19.8	74.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	45.9	16.1	23.4	35.2	49.1	59.5	68.0	71.9	68.5	59.1	46.9	32.0	19.5
	DBStd	20.72	8.15	9.09	9.70	9.24	7.36	5.60	4.73	4.99	6.54	7.79	8.58	8.07
	HDD50	4044	1050	746	466	126	13	0	0	0	9	149	540	945
	HDD65	7497	1515	1166	923	481	197	34	4	24	192	561	990	1410
	CDD50	2543	0	0	8	99	308	540	679	572	284	53	0	0
	CDD65	522	0	0	0	3	28	124	218	132	17	0	0	0
	CDH74	3470	0	0	0	32	222	878	1556	695	87	0	0	0
	CDH80	744	0	0	0	4	26	179	411	109	15	0	0	0
Wind	WSAvg	6.3	5.7	6.0	6.9	7.3	7.0	6.4	6.0	5.8	5.8	5.9	6.3	6.0
Precipitation	PrecAvg	15.5	0.1	0.2	0.4	0.6	1.1	1.6	3.8	4.7	1.9	0.9	0.2	0.0
	PrecMax	28.0	0.5	1.0	1.8	3.3	4.3	4.8	7.5	9.6	5.2	2.0	1.1	0.2
	PrecMin	7.8	0.0	0.0	0.0	0.0	0.0	0.4	1.0	0.7	0.5	0.1	0.0	0.0
	PrecStd	4.6	0.1	0.2	0.4	0.7	0.9	1.1	2.0	2.4	1.1	0.6	0.3	0.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.2	49.8	65.7	78.1	83.3	87.8	90.5	85.9	82.1	70.6	56.7	41.1
		MCWB	29.4	34.8	43.0	50.5	53.7	57.6	64.2	61.3	56.5	51.0	40.0	30.2
	2%	DB	35.3	44.1	58.6	72.1	79.3	84.4	86.9	83.1	75.9	65.8	51.7	36.6
		MCWB	25.9	31.2	40.1	48.2	52.4	56.8	61.8	62.0	55.4	48.7	37.5	27.5
	5%	DB	31.0	39.9	54.0	67.9	76.3	81.8	84.4	80.5	72.8	62.3	47.9	33.2
		MCWB	23.5	29.5	37.6	46.0	51.4	56.1	61.2	61.2	54.5	46.6	36.0	25.4
	10%	DB	27.7	36.5	49.7	63.9	73.0	79.2	81.9	78.0	69.8	58.8	44.2	30.2
		MCWB	21.2	27.4	35.3	44.0	49.7	55.9	61.0	60.5	53.9	45.1	34.5	23.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	29.9	35.8	45.3	54.7	57.8	64.2	70.0	69.6	63.5	54.5	43.2	31.4
		MCDB	38.5	46.3	61.2	68.3	70.5	74.8	80.1	78.3	68.1	64.4	49.9	37.7
	2%	WB	26.5	32.6	41.3	51.0	55.7	62.0	68.0	67.4	60.8	51.7	40.1	28.2
		MCDB	34.9	41.1	56.1	65.3	70.2	72.2	77.5	75.4	68.8	60.5	47.9	35.1
	5%	WB	23.8	30.3	38.6	48.4	54.1	60.6	66.4	65.8	58.3	49.5	37.6	26.1
		MCDB	30.3	39.0	52.1	63.3	68.8	72.4	76.2	73.2	66.3	57.9	45.0	31.9
	10%	WB	21.4	28.0	36.0	45.9	52.5	59.3	64.9	64.3	56.4	47.5	35.2	24.2
		MCDB	27.4	35.8	47.9	60.4	67.7	72.1	75.0	72.2	65.8	55.5	43.5	30.1
Mean Daily Temperature Range	5% DB	MDBR	14.6	15.8	17.1	18.1	18.3	17.2	15.9	14.9	15.1	15.5	14.4	13.4
		MCDBR	17.7	18.8	21.1	22.1	21.6	20.5	19.0	17.7	18.7	19.1	17.9	16.0
		MCWBR	12.2	11.5	11.6	10.6	9.2	7.0	6.0	6.0	7.2	8.9	10.0	11.0
	5% WB	MCDBR	16.9	17.1	19.9	19.3	18.6	16.7	14.7	14.0	14.3	16.0	15.3	14.5
		MCWBR	12.0	10.9	11.7	10.3	8.8	7.0	6.4	6.7	7.3	8.7	9.8	10.4
Clear-Sky Solar Irradiance	taub	0.312	0.370	0.403	0.451	0.481	0.531	0.502	0.506	0.473	0.412	0.344	0.316	
	taud	2.179	2.044	1.973	1.908	1.890	1.829	1.972	1.936	1.953	2.036	2.151	2.188	
	Ebn at Noon	265	261	266	261	256	243	249	243	241	243	248	252	
	Edh at Noon	33	43	52	59	62	66	57	57	53	43	33	30	
All-Sky Solar Radiation	RadAvg	859	1145	1584	1928	2132	2036	1930	1740	1476	1235	922	756	
	RadStd	73	84	130	126	123	142	88	100	96	81	101	72	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.79	N/A	-2.56	N/A	N/A	N/A	N/A	-270	N/A	N/A	
(47)	Regional (4 neighbors)	N/A	N/A	-2.05	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page